

S.A. ENGINEERING COLLEGE

Autonomous Institution | Affiliated to Anna University | NBA & NAAC 'A' Grade

Dept of Electronics & Communication Engg & Dept of Electronics Engg [VLSI Design & Technology]

KERNEL PRIME '26 - SOFTWARE TRACK

National Level 24-Hour Hackathon | October 8-9, 2026

OFFICIAL PROBLEM STATEMENT: PS-01

Decentralized Supply Chain Provenance & Traceability Engine

Domain: Web3 / Distributed Systems / Cryptography

1. PROBLEM OVERVIEW & CHALLENGE STATEMENT

Architect a verifiable cryptographic ledger system to eliminate counterfeits, trace raw material custody across multi-tier supplier networks, and produce real-time ESG compliance proofs.

2. MANDATORY ARCHITECTURAL REQUIREMENTS

[REQ-1] Multi-party state machine with cryptographic signatures for handover validation.

[REQ-2] Decentralized identity (DID) verification for authorized suppliers and logistics carriers.

[REQ-3] Immutable audit trail with zero-knowledge proof verification for confidential bill-of-materials

[REQ-4] Real-time automated dispute arbitration and smart contract escrow settlement.

[REQ-5] Sub-second query latency for public consumer barcode/QR verification portals.

3. HACKATHON EVALUATION MATRIX

* System Architecture & Concurrency Resilience: 25%

* Technical Innovation & Production Viability: 25%

* Live Demo & Real-World Latency Benchmarks: 30%

* Security Hardening, UI Polish & Documentation: 20%

CONFIDENTIAL - AUTHORIZED ALLOCATION COPY ONLY

Generated securely via Kernel Prime Allocation Engine. Strictly non-transferable.